

Jacobi Iterative Method

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$$\mathbf{Ax} = \mathbf{b}$$

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2$$

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$$a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \cdots + a_{nn}x_n = b_n$$

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Diagonally Dominant System:

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$$|a_{ii}| \geq \sum_{j \neq i} |a_{ij}|$$

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Example 03-05: Find root of the following system of equations using Jacobi iterative method with initial guess $\mathbf{x}^{(0)} = \mathbf{1}$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$\begin{array}{rcccccccl} 10x_1 & - & x_2 & + & 2x_3 & & & = & 6 \\ - & x_1 & + & 11x_2 & - & x_3 & + & 3x_4 & = & 25 \\ & 2x_1 & - & x_2 & + & 10x_3 & - & x_4 & = & -11 \\ & & & 3x_2 & - & x_3 & + & 8x_4 & = & 15 \end{array}$$

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$$|10| > |-1| + |2| + |0| = 3$$

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$\mathbf{A}$ is diagonally dominant matrix.

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$$x_1^{(k)} = \frac{1}{10} \cdot [6 - (-x_2^{(k-1)}) - 2x_3^{(k-1)}]$$

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$$x_1^{(k)} = \frac{1}{10} \cdot [6 - (-x_2^{(k-1)}) - 2x_3^{(k-1)}]$$
$$x_2^{(k)} = \frac{1}{11} \cdot [25 - (-x_1^{(k-1)}) - (-x_3^{(k-1)}) - 3x_4^{(k-1)}]$$

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$$\begin{array}{ccccccc} & 10x_1 & - & x_2 & + & 2x_3 & = & 6 \\ - & x_1 & + & 11x_2 & - & x_3 & + & 3x_4 & = & 25 \\ & 2x_1 & - & x_2 & + & 10x_3 & - & x_4 & = & -11 \\ & & & 3x_2 & - & x_3 & + & 8x_4 & = & 15 \end{array}$$

Recurrence Relations:

$$x_1^{(k)} = \frac{1}{10} \cdot [6 - (-x_2^{(k-1)}) - 2x_3^{(k-1)}]$$

$$x_2^{(k)} = \frac{1}{11} \cdot [25 - (-x_1^{(k-1)}) - (-x_3^{(k-1)}) - 3x_4^{(k-1)}]$$

$$x_3^{(k)} = \frac{1}{10} \cdot [-11 - 2x_1^{(k-1)} - (-x_2^{(k-1)}) - (-x_4^{(k-1)})]$$

Jacobi Iterative Method

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Jacobi Iterative Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

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$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = \frac{1}{10} \cdot [6 - (-1) - 2 \cdot 1]$$

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$$k = 1$$

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$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

$$x_1^{(1)} = 0.5$$

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 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.181818, -1.1, 1.625)$$

Jacobi Iterative Method

 $k = 1$

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$$x_1^{(2)} = \frac{1}{10} \cdot [6 - (-x_2^{(1)}) - 2x_3^{(1)}]$$

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Jacobi Iterative Method

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$$x_2^{(2)} = \frac{1}{11} \cdot [25 - (-x_1^{(1)}) - (-x_3^{(1)}) - 3x_4^{(1)}]$$

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$$x_2^{(2)} = \frac{1}{11} \cdot [25 - (-0.5) - (-(-1.1)) - 3 \cdot 1.625]$$

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 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.181818, -1.1, 1.625)$$

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Jacobi Iterative Method

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$k = 2$

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Jacobi Iterative Method

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Jacobi Iterative Method

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 $k = 2$

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$$x_1^{(2)} = 1.038182$$

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$$x_4^{(2)} = \frac{1}{8} \cdot [15 - 3x_2^{(0)} - (-x_3^{(0)})]$$

Jacobi Iterative Method

 $k = 1$

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 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.181818, -1.1, 1.625)$$

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Jacobi Iterative Method

$k = 1$

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$$x_2^{(2)} = 1.775$$

$$x_3^{(2)} = -0.819318$$

$$x_4^{(2)} = 0.919318$$

Jacobi Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.181818$$

$$x_3^{(1)} = -1.1$$

$$x_4^{(1)} = 1.625$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.181818)^2 + (-1.1)^2 + (1.625)^2} = 2.976736$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.181818)^2 + (-2.1)^2 + (0.625)^2} = 2.539157$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.539157}{2.976736} = 0.853001 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.181818, -1.1, 1.625)$$

$$x_1^{(2)} = 1.038182$$

$$x_2^{(2)} = 1.775$$

$$x_3^{(2)} = -0.819318$$

$$x_4^{(2)} = 0.919318$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(1.038182)^2 + (1.775)^2 + (-0.819318)^2 + (0.919318)^2}$$

Jacobi Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.181818$$

$$x_3^{(1)} = -1.1$$

$$x_4^{(1)} = 1.625$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.181818)^2 + (-1.1)^2 + (1.625)^2} = 2.976736$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.181818)^2 + (-2.1)^2 + (0.625)^2} = 2.539157$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.539157}{2.976736} = 0.853001 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.181818, -1.1, 1.625)$$

$$x_1^{(2)} = 1.038182$$

$$x_2^{(2)} = 1.775$$

$$x_3^{(2)} = -0.819318$$

$$x_4^{(2)} = 0.919318$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(1.038182)^2 + (1.775)^2 + (-0.819318)^2 + (0.919318)^2} = 2.396847$$

Jacobi Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.181818$$

$$x_3^{(1)} = -1.1$$

$$x_4^{(1)} = 1.625$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.181818)^2 + (-1.1)^2 + (1.625)^2} = 2.976736$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.181818)^2 + (-2.1)^2 + (0.625)^2} = 2.539157$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.539157}{2.976736} = 0.853001 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.181818, -1.1, 1.625)$$

$$x_1^{(2)} = 1.038182$$

$$x_2^{(2)} = 1.775$$

$$x_3^{(2)} = -0.819318$$

$$x_4^{(2)} = 0.919318$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(1.038182)^2 + (1.775)^2 + (-0.819318)^2 + (0.919318)^2} = 2.396847$$

$$\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2 = \sqrt{(0.538182)^2 + (-0.406818)^2 + (0.280682)^2 + (-0.705682)^2}$$

Jacobi Iterative Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.181818$$

$$x_3^{(1)} = -1.1$$

$$x_4^{(1)} = 1.625$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.181818)^2 + (-1.1)^2 + (1.625)^2} = 2.976736$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.181818)^2 + (-2.1)^2 + (0.625)^2} = 2.539157$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.539157}{2.976736} = 0.853001 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.181818, -1.1, 1.625)$$

$$x_1^{(2)} = 1.038182$$

$$x_2^{(2)} = 1.775$$

$$x_3^{(2)} = -0.819318$$

$$x_4^{(2)} = 0.919318$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(1.038182)^2 + (1.775)^2 + (-0.819318)^2 + (0.919318)^2} = 2.396847$$

$$\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2 = \sqrt{(0.538182)^2 + (-0.406818)^2 + (0.280682)^2 + (-0.705682)^2} = 1.015830$$

Jacobi Iterative Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.181818$$

$$x_3^{(1)} = -1.1$$

$$x_4^{(1)} = 1.625$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.181818)^2 + (-1.1)^2 + (1.625)^2} = 2.976736$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.181818)^2 + (-2.1)^2 + (0.625)^2} = 2.539157$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.539157}{2.976736} = 0.853001 > \varepsilon_t \rightarrow \text{continue}$$

 $k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.181818, -1.1, 1.625)$$

$$x_1^{(2)} = 1.038182$$

$$x_2^{(2)} = 1.775$$

$$x_3^{(2)} = -0.819318$$

$$x_4^{(2)} = 0.919318$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(1.038182)^2 + (1.775)^2 + (-0.819318)^2 + (0.919318)^2} = 2.396847$$

$$\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2 = \sqrt{(0.538182)^2 + (-0.406818)^2 + (0.280682)^2 + (-0.705682)^2} = 1.015830$$

$$\varepsilon^{(2)} = \frac{\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2}{\|\mathbf{x}^{(2)}\|_2} = \frac{1.015830}{2.396847} = 0.423819 > \varepsilon_t$$

Jacobi Iterative Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$x_1^{(1)} = 0.5$$

$$x_2^{(1)} = 2.181818$$

$$x_3^{(1)} = -1.1$$

$$x_4^{(1)} = 1.625$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.5)^2 + (2.181818)^2 + (-1.1)^2 + (1.625)^2} = 2.976736$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.5)^2 + (1.181818)^2 + (-2.1)^2 + (0.625)^2} = 2.539157$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.539157}{2.976736} = 0.853001 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.5, 2.181818, -1.1, 1.625)$$

$$x_1^{(2)} = 1.038182$$

$$x_2^{(2)} = 1.775$$

$$x_3^{(2)} = -0.819318$$

$$x_4^{(2)} = 0.919318$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(1.038182)^2 + (1.775)^2 + (-0.819318)^2 + (0.919318)^2} = 2.396847$$

$$\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2 = \sqrt{(0.538182)^2 + (-0.406818)^2 + (0.280682)^2 + (-0.705682)^2} = 1.015830$$

$$\varepsilon^{(2)} = \frac{\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2}{\|\mathbf{x}^{(2)}\|_2} = \frac{1.015830}{2.396847} = 0.423819 > \varepsilon_t \rightarrow \text{continue}$$